

Drawing the burn-scar labels

The project file is `data/labels/biak_labels.qgz`. Open it in QGIS 3.44 LTR.

These polygons become the reference that the Task 18 burned-area product is measured against (Task 22). Nothing else in this project can do that: the field survey was stopped by security conditions, and F18 showed that no prompted segmentation model can stand in.

What you are looking at

layer	what it is
LABELS — draw here	empty; your polygons go here
samples	42 points, 7 inside and 7 outside each event's VIIRS footprint
VIIRS footprints	375 m circles round the detections — where the samples were drawn from, not truth
VIIRS/MODIS detections	the individual satellite detections
S2DR4 1 m imagery	SWIR and true colour for each event, 28 August 2026

Only the SWIR image for E0313 is switched on at the start. Tick the others as you move between events.

There is no dNBR+ layer, on purpose. These labels judge that product, so seeing it while drawing would make the comparison worthless. Do not add it.

The imagery

SWIR (B12/B8A/B4) is the one to judge from. In it:

what you see	what it is
bright green	healthy vegetation, the greener the denser
brown to maroon, dull red	burnt or bare ground — this is what you are looking for
pale green to cream	grass, dry or sparse cover
near black	water
white, sharp edges	roofs and paved surfaces

Switch to true colour when you want to check what a feature is — a roof, a track, a bare field — because true colour looks like a photograph and SWIR does not.

The three channels share one fixed brightness scale, so these colours mean the same thing

in every tile and can be compared between events. (An earlier version of this imagery stretched each channel separately, which showed vegetation orange and burnt ground blue; if you see that, the imagery is stale — re-run `scripts/label_prepare.py`.)

The 1 m detail is super-resolved, which means a model generated it from 10 m pixels. Trust the shape of a patch and the contrast between patches. Do not trust a single sharp edge or a small isolated object: those are the parts a super-resolution model invents.

The job: judge 42 boxes

There is nothing to draw. The label layer already holds **42 boxes of 60 x 60 m**, one centred on each sample point, 0.36 ha each, which is 9 pixels of the 20 m product being tested. Your job is the judgement only: what is inside each box.

Why fixed boxes rather than outlines: the sample location has to come from the random draw.

Choosing where to draw, by eye, around a scar that is visible turns the reference into a collection of obvious cases, and the agreement figure then measures the easy ones only.

1. Open the project, select **LABELS — draw here**, open the attribute table (**F6**).
2. Toggle editing (**Ctrl+E**).
3. Click a row, then the **Zoom to feature** button, so the map shows that box.
4. Fill two columns:
5. `class` — `burned`, `unburned` or `unsure`;
6. `confidence` — `high`, `medium` or `low`;
7. `notes` — only if something needs saying.
8. Save (**Ctrl+S**) often.

The table also has a **form view** at the bottom right, which shows one box at a time if that reads more comfortably than the grid.

Spelling matters, and `scripts/label_check.py` catches anything unexpected.

If a box straddles two cover types

Judge what covers most of it. If it is close to half and half, that is exactly what `unsure` is for.

The rules that keep this honest

- `unsure` is a real answer. Thin smoke, cloud shadow, a patch you would argue about —
- mark it `unsure` and move on. Task 22 drops those and reports how many there were. A
- guessed label is worse than no label, because it silently moves the accuracy figure.

- **Do not move a box** to somewhere easier to judge, and do not resize it. The points were
- drawn at random from a fixed seed; shifting them onto convenient ground is exactly the
- bias the random draw exists to prevent. A box on water, on a roof or under cloud gets
- `unsure` and a note.
- **Inside the footprint does not mean burned.** The footprint only says a satellite saw
- heat within 375 m at some point. Plenty of ground inside it did not burn, and that is
- precisely what the labels are meant to establish. Judge each patch on the imagery.
- **Outside the footprint does not mean unburned** either. If it looks burned, label it
- burned.
- **Do not use the detections as evidence.** They tell you where heat was seen, which is
- the very thing the product is trying to convert into area. Use them to orient yourself,
- not to decide.

When you are done

```
python scripts/label_check.py
```

It reports how many polygons exist, whether each holds its own sample point, the class counts by stratum, the polygon sizes, and anything malformed or misspelled. It changes nothing.

Partial work is fine — the check tells you what is left. Task 22 can run on fewer than 42 polygons, it will just say so and carry a wider uncertainty.

The three scripts, in the order you need them

script	when	what it does
<code>scripts/label_prepare.py</code>	already run	builds the imagery, the sample points and the empty label layer
<code>scripts/label_fill_ids.py</code>	only if you add polygons by hand	fills <code>sample_id</code> and <code>event_id</code> from the point inside each polygon
<code>scripts/label_check.py</code>	any time	reports what is there and what is wrong

`scripts/label_qgis_project.py` rebuilds the `.qgz` itself. It never touches the GeoPackage, so polygons already drawn survive a rebuild.